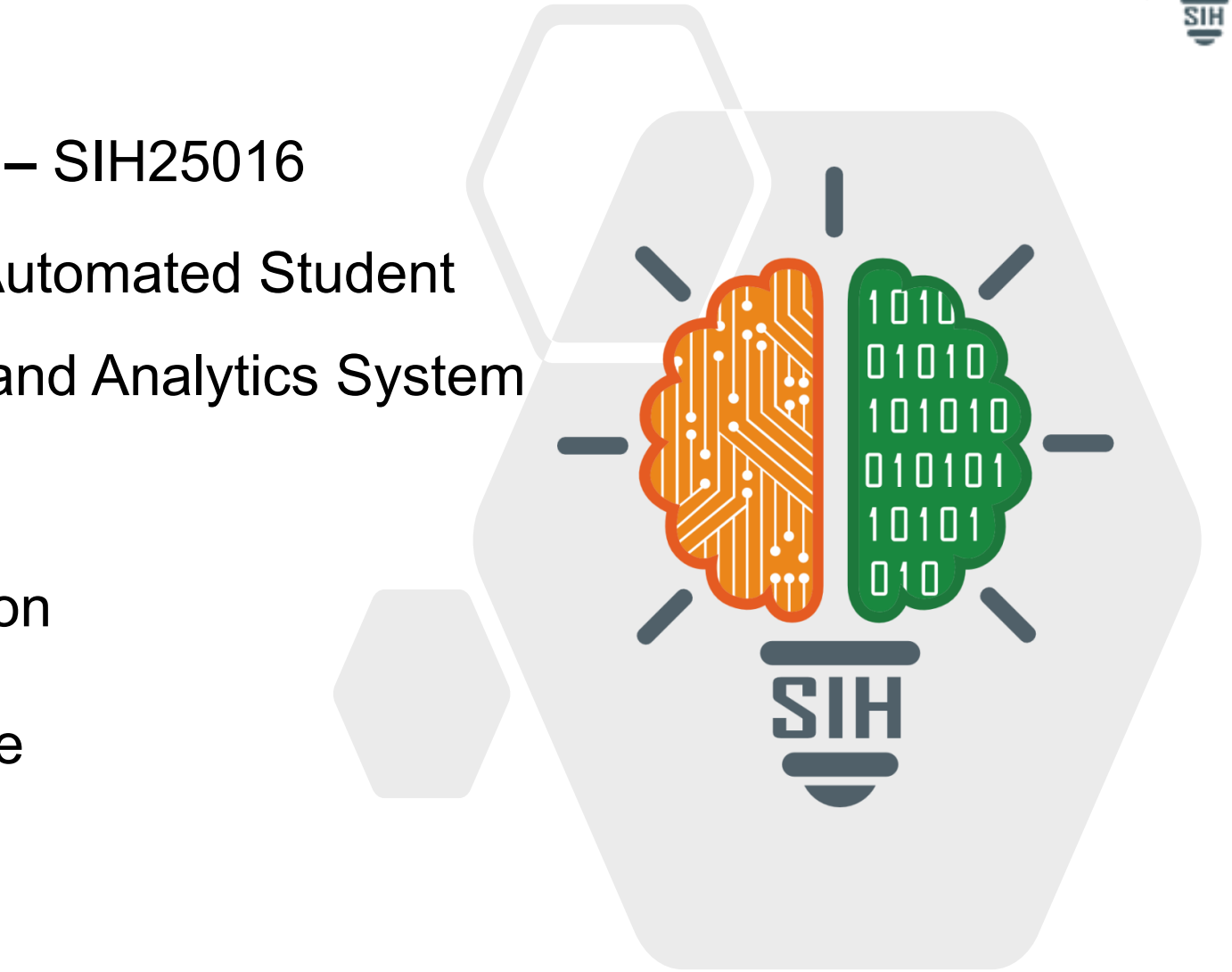
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- **Problem Statement ID –** SIH25016
- **Problem Statement -** Automated Student Attendance Monitoring and Analytics System for Colleges
- **Theme -** Smart Education
- **PS Category –** Software
- **Team ID -** From portal
- **Team Name –** AttendSmart



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Detailed explanation of proposed solution

Problem Statement:

- Large classes → harder to track
- Manual roll calls & paper registers waste time
- Errors & proxy attendance can occur
- No easy insights for faculty/admin
- Outdated system slows digital transformation

Proposed Solution:

We propose a proxy-proof attendance system that uses a mobile app for multi-factor verification: face liveness, BLE proximity, and barometer checks. It features a live teacher dashboard, operates 100% offline to ensure privacy and accuracy, and completely eliminates proxy attendance

Our Project Promises:

- Saves time
- No proxy or errors
- Secure, offline-first with local biometric handling
- Instant teacher insights
- Easy to use



Authentications

Face Recognition
Uses facial detection with liveness checks

Location Verification
Multi-layered GPS, BLE, and barometric sensors

→ Primary

Fingerprint
Alternative biometric method

QR Code
Time-sensitive code for fallback

Peer-to-Peer Verification
Random classmate confirmation (when flagged)

Manual Verification
Teacher-logged attendance

→ Secondary Fallbacks

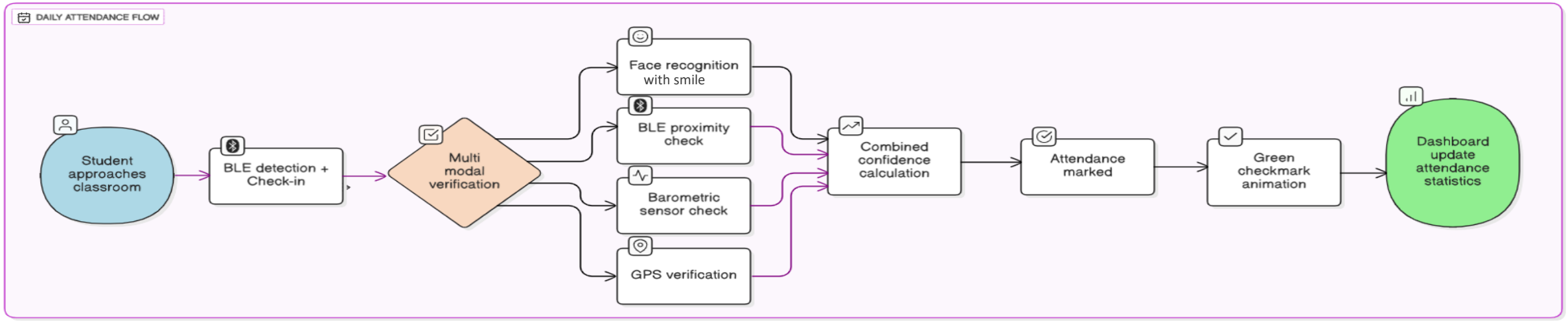
For better understanding watch this video

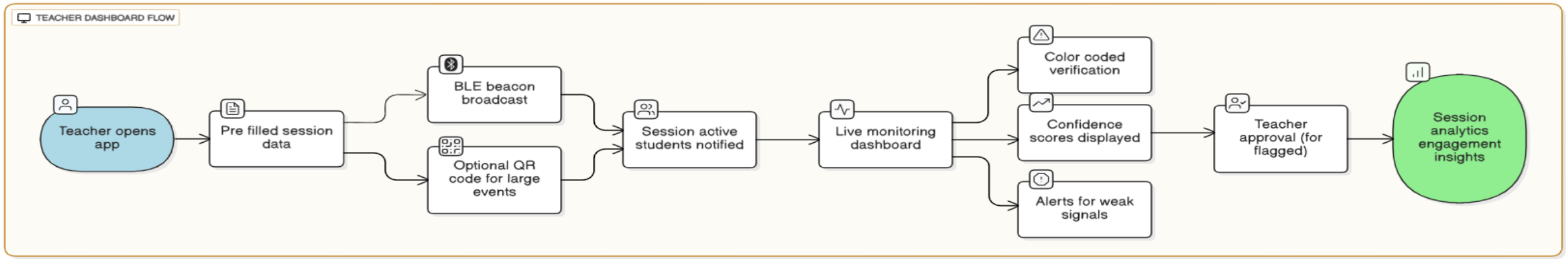
https://youtube.com/playlist?list=PLZghJ684ztTeF9GQ2ZqvPMktYVlb5e7lk&si=yODhyAmdDXDmYw-V

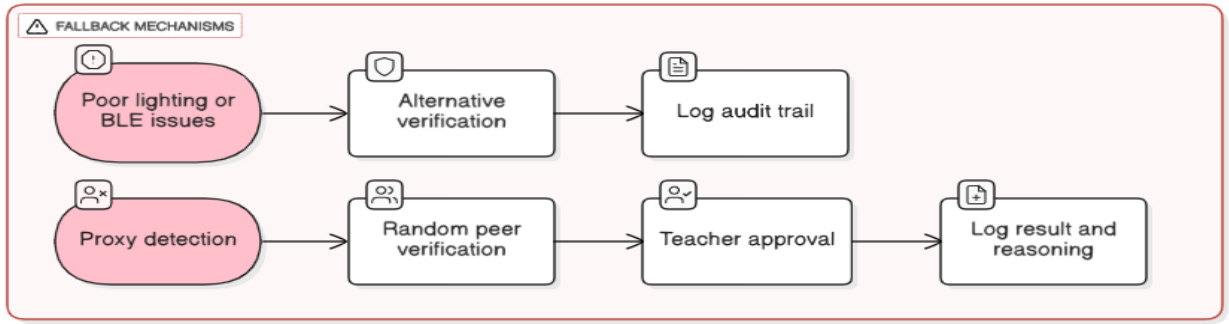
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TECHNICAL APPROACH

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Check out our Prototype Demo Video

https://youtube.com/playlist?list=PLZghJ684ztTeF9GQ2ZqvPMktYVlb5e7lk&si=yODhyAmdDXDmYw-V

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea

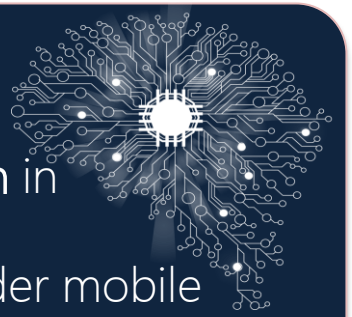








Potential Challenges:



- Accuracy of biometric **recognition** in
- **varying lighting/conditions**
- Connectivity/battery issues on older mobile devices
- **Privacy** or resistance to on-device biometrics
- **Scaling** and **managing data** for thousands or millions of students

Strategies to overcome:

- Use proven face recognition models and liveness checks for reliability
- **Offline**-first app and **battery optimizations** for minimal power/network use
- All biometric processing on device; no cloud storage; **transparent data policy**
- Modular, cloud-compatible backend for large-scale deployment
- Continuous UX feedback loop with students and teachers

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IMPACT AND BENEFITS

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Impact:

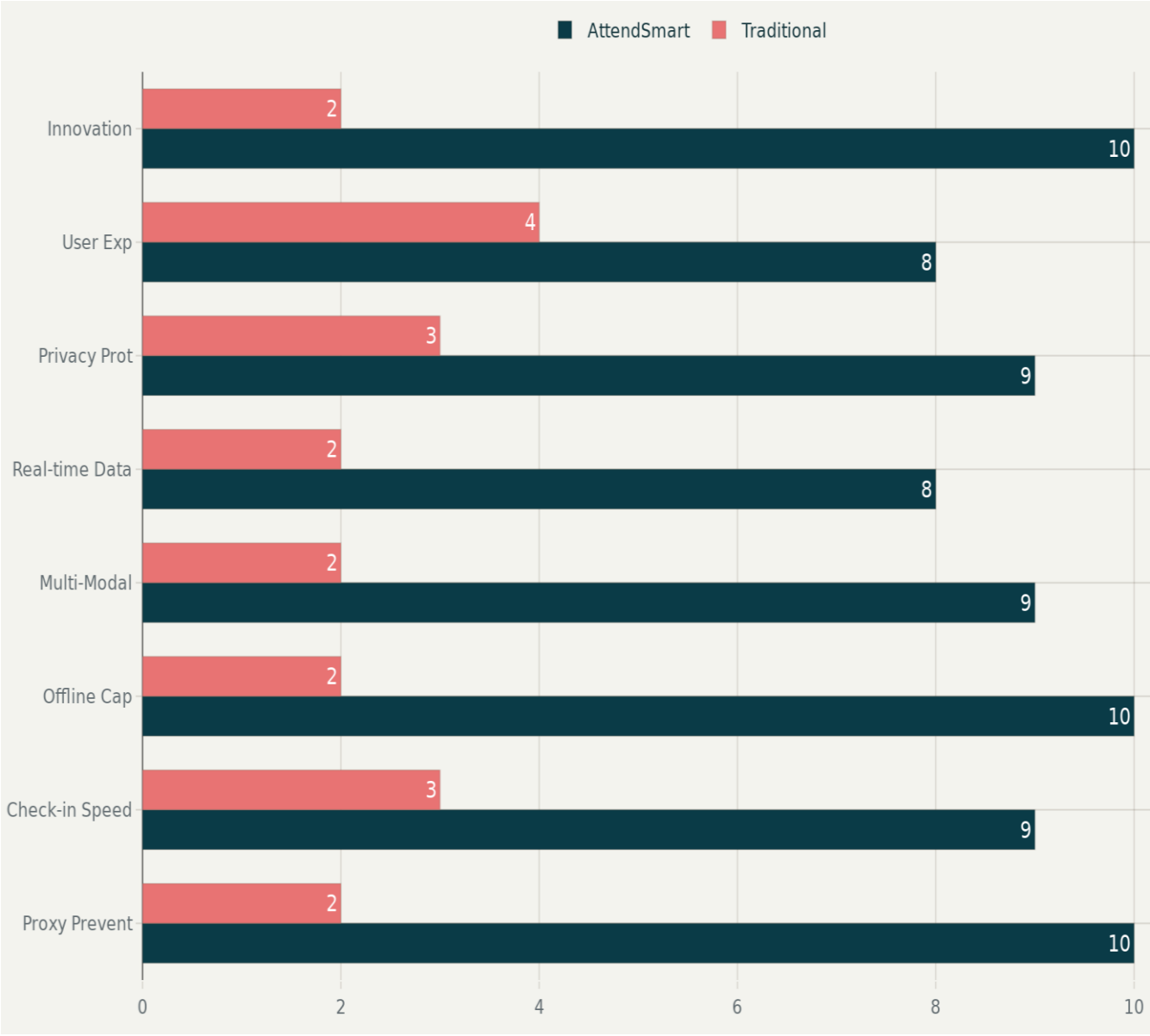
- **Proxy-proof** attendance: Face + BLE + GPS + Barometer.
- **3–5s check-in**; saves 15+ min/class.
- 100% offline; on-device biometrics

Analogy:

- Manual gate → e-gate.
- 2FA for attendance: “who” + “where.”

Benefits:

- **Efficiency**: 3–5s per student; auto session detection
- **Accuracy**: 95%+ with liveness; <0.1% false positives
- **Scale & privacy**: offline-first, state-ready, data stays on device.



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RESEARCH AND REFERENCES

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Reference	Technology Focus	AttendSmart Connection	Citations
[1] N. Surantha et al., 'Lightweight face recognition-based portable attendance system with anti-spoofing,' ScienceDirect, 2024	Face Recognition + Anti-Spoofing	Liveness detection (smile/blink) to prevent photo spoofing	22 citations
[2] H. Xia et al., 'Using Multiple Barometers to Detect Floor Location of Smart Phones,' PMC, 2015	Barometric Floor Positioning	Floor-level verification using smartphone sensors	136 citations
[3] R. Lodha et al., 'Bluetooth Smart Based Attendance Management System,' Procedia Computer Science, 2015	BLE Proximity Detection	8-meter proximity verification using Bluetooth Low Energy	200+ citations
[4] D. Saraswat et al., 'Anti-spoofing-enabled Contactless Attendance Monitoring System,' PMC, 2023	Contactless Anti-Spoofing	Proxy prevention through contactless biometric verification	20 citations
[5] J. Samatha et al., 'Smart attendance monitoring system using multimodal biometric authentication,' 2025	Multi-Modal Integration	Multiple verification layers for enhanced security	Recent publication